

Engineering Empathy: A Blueprint for Interdisciplinary Art and Engineering in K-12 Education

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Figure 1: Undergraduate Scholar Leads Game Development Activity with K-12 Students

Abstract

The Limbitless ENGAGE program at the University of Central Florida (UCF) addresses the fragmentation of STEM education by introducing a scalable framework that synthesizes art, game design, and engineering through human-centric bionic design. While many STEAM initiatives struggle to bridge disciplinary silos, this program utilizes artistic bionic-arm design as a "human-centric hook" to foster genuine interdisciplinary collaboration among K-12 students and undergraduate mentors.

Participants engage in modules spanning biomedical engineering, real-time rendering, and interactive logic. Key activities include using visual storytelling to create training games for bionic users, effectively leveraging game design as a functional tool for rehabilitation and empowerment. The program's "Near-Peer" mentorship

model supported by a cohort of over 70 interdisciplinary undergraduate scholars, provides a unique pipeline that connects high-level university research with K-12 outreach.

For the SIGGRAPH community, Limbitless ENGAGE contributes a significant innovation in "Graphics for Social Impact." It demonstrates how interactive systems and real-time rendering can transcend entertainment, serving as critical components in empathy-driven education and medical simulation. The primary takeaway is a replicable "Bionic Blueprint," a modular curriculum that treats social impact as a core metric for success. By merging technical rigor with creative expression, the program empowers students to master complex digital tools while maintaining a grounded focus on real-world service. This submission offers a road map for educators and technologists to architect inclusive, interdisciplinary environments that inspire the next generation of creators to design for humanity.



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CCS Concepts

• Computing methodologies → Graphics systems and interfaces; Shape modeling; Animation; • Human-centered computing → Accessibility systems and tools; Accessibility design and evaluation methods; Interaction design theory, concepts and paradigms.

Keywords

Adaptive or Assistive Technology, Design, Education, Fabrication/Manufacturing, Games, Game Design/Development, Robotics

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1 OVERVIEW

The ENGAGE program is an implementation of a multi-modal educational framework designed to bridge the gap between creative art and engineering. On average, each semester brings together more than 70 interdisciplinary scholars from six UCF colleges to drive this mission of providing cost-free bionic devices to children with limb differences. These scholars, who represent fields such as fine arts, game design, mechanical engineering, and health sciences, are the foundation of the ENGAGE program's K-12 outreach. By describing their professional practice in a simplified way, undergraduate scholars mentor younger students, helping them understand the essential role of creativity and digital art in problem-solving through immersive, guided STEAM experiences. In doing so, scholars highlight a unified "Blueprint of Art and Engineering" long before the K-12 student participants pursue higher education.

1.1 The Problem: Siloed Technical Literacy

Most students are not exposed to the synthesis of art and engineering until college. Traditional K-12 STEM programs often prioritize functional outcomes while ignoring the aesthetic intent and user experience that are essential to human-centered technology.

1.2 Limbless Solutions' ENGAGE Educational Model

Limbless Solutions addresses complex medical and accessibility challenges built on a "Near-Peer" mentoring model[Jett et al. 2005] driven by visual storytelling and creative art. This approach integrates undergraduate scholars into K-12 education, where they translate advanced professional workflows, including digital sculpting, game design, and training, 3D printing preparation, and game engine integration into engaging, hands-on modules.

2 THE BIONIC BLUEPRINT: THREE TRACKS OF ENGAGEMENT

The ENGAGE framework is built upon three primary tracks that mirror professional pipelines in graphics and engineering, adapted for a K-12 audience through the "Near-Peer" mentorship[Jett et al. 2005] of undergraduate scholars.

Track 1: Digital-to-Physical Aesthetics (Sleeve Sculpting)

In this track, students transition from passive consumers of technology to active creators of identity through mentorship by undergraduate facilitators. Utilizing professional 3D modeling software, such as Autodesk Maya or ZBrush, undergraduate students apply their foundational knowledge of mesh manipulation and topology to teach K-12 students how to design bionic sleeves, which are

removable, aesthetic covers for prosthetic arms. The Hook: By focusing on personalization, students explore how texture mapping, color theory, and shape language influence the user's psychology and technology adoption. The Workflow: Undergraduate students practice digital sculpting and prepare their models for 3D printing, learning how to navigate the constraints of physical materials along with the creative freedom of digital media. Through this process, they develop advanced design and technical skills that they translate into accessible and engaging K-12 learning experiences.

Track 2: Interactive Medical Training (Video Game Concepting) This track re-contextualizes game design as a functional tool for therapeutic intervention. Undergraduate students utilize real-time engines (such as Unity or Unreal Engine) to conceptualize and prototype game mechanics that respond to EMG (electromyography) sensors. The Objective: Rather than designing solely for entertainment, undergraduate students develop "training games" that support children with limb differences utilize their bionic arm through game-play. Graphics for Good: This highlights how interactive logic and real-time rendering are critical in medical rehabilitation, requiring students to solve complex problems regarding user feedback and interactive systems. They then distill these concepts in game development and make ideas human-computer interaction approachable for younger audiences.

Track 3: Mechanical Logic (The Bionic Model) Undergraduates explore the relationship between digital animation and physical kinesthetics. By building physical models that replicate the "rigging" found in 3D animation software, they learn how animation constraints translate to mechanical limits. Interdisciplinary Link: This track bridges the gap between the virtual "bones" of a character rig and the mechanical pivots of a prosthetic arm. Outcome: Through the ENGAGE program, undergraduates support the K-12 students to gain a deeper understanding of how rigging logic informs the engineering of wearable technology, emphasizing that a successful bionic design requires a seamless blend of mechanical function and animated movement.

3 IMPACT AND ASSESSMENT: FROM "HOW" TO "WHY"

The success of the ENGAGE model is measured not just by technical proficiency, but by a fundamental shift in student perspective regarding the purpose of technology.

3.1 Student Transformation and Integrated Iteration

Most K-12 technical curricula focus on the "how," the rote mechanics of coding or CAD. ENGAGE introduces the "why" through the lens of empathy. When a student recognizes that their artistic choice affects a user's confidence or their game mechanic improves a user's muscle control, they engage in Integrated Iteration. In this state, engineering is no longer a separate task; it becomes a necessary vehicle to support human-centric artistic needs.

3.2 Quantifiable Success And Replicability

Scalability: The framework successfully manages over 70 interdisciplinary undergraduate scholars from six different university colleges (including Arts and Humanities, Engineering, and Health

Sciences). This demonstrates that building on the "Near-Peer" model is a functional approach that promotes collaboration between colleges and promotes large-scale university-to-community outreach.

Measured Confidence: For measurement purposes, a framework including the S-STEM Survey [Unfried et al. 2015] and the RIS-STEM [Paul et al. 2020] is proposed to evaluate pre and post impressions on student attitudes in STEM. The hypothesis is that the interventions will influence participants for a shift in viewing their skills as part of a unified problem-solving toolkit.

4 CONCLUSION

The Limbitless ENGAGE program demonstrates that when we dismantle the silos between art and engineering, we unlock a more profound level of student engagement. By positioning Computer Graphics as the bridge between technical rigor and human empathy, we empower the next generation of creators to view their work as a tool for social good. For the SIGGRAPH community, this program serves as both a road map and a challenge: to move beyond technical mastery for its own sake and toward a pedagogical model that inspires students to design for humanity. The ENGAGE "Bionic

Blueprint" intends to offer a replicable, scalable, and deeply meaningful framework that prepares students to lead in an increasingly interdisciplinary world.

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